TryHackMe - Windows Basics

Platform: TryHackMe
Room link: Windows Basics

1. Introduction

Windows is one of the most widely used operating systems in the world. It provides a graphical user interface (GUI) that allows users to interact with computers, manage files, run applications, and configure system settings.

For cybersecurity professionals, understanding Windows is important because many organizations use Windows-based computers, servers, and enterprise environments.

This room introduces the Windows operating system, its desktop environment, file management, system utilities, and basic administration.

2. Objectives

The main objectives of this room are:

- Understand the Windows operating system.
- Explore the Windows desktop environment.
- Understand the Start Menu and Taskbar.
- Learn how to navigate File Explorer.
- Understand Windows files and folders.
- Explore Windows system settings.
- Learn about user accounts and permissions.
- Understand basic Windows administration tools.
- Become familiar with Windows security features.

3. Tools and Requirements

Tool	Purpose
Windows VM	Practical Windows environment
File Explorer	File and folder management
Task Manager	Process and resource monitoring
Control Panel	System configuration
Settings	Windows configuration
Command Prompt	Command-line operations
PowerShell	Windows administration and automation

4. Task-wise Walkthrough

Task 1 - Introduction to Windows

Microsoft Windows is a family of operating systems developed by Microsoft.

It provides an environment for running applications, managing hardware, accessing files, and connecting to networks.

Windows is commonly used in:

- Personal computers.
- Business workstations.
- Enterprise environments.
- Servers.
- Virtual machines.

Some commonly used Windows versions include:

- Windows 7
- Windows 8
- Windows 10
- Windows 11
- Windows Server editions

Understanding the Windows environment is an important foundation for learning system administration and cybersecurity.

Task 2 - Windows Desktop Environment

The Windows desktop is the primary graphical workspace users interact with after signing in.

It contains several important components.

i. **Desktop**: The desktop is the main workspace where users can place shortcuts, files, and folders.
ii. **Start Menu**: The Start Menu provides access to installed applications, system settings, search, and power options.
iii. **Taskbar**: The Taskbar displays running applications, pinned programs, system notifications, and other shortcuts.
iv. **Notification Area**: The notification area displays background application icons and system information, such as network connectivity, sound, and battery status.
v. **Search**: Windows Search allows users to locate applications, files, settings, and other system resources.

Task 3 - Start Menu and Taskbar

The Start Menu is one of the primary navigation components in Windows.

It provides access to:

- Installed applications.
- Recently used files.
- Search functionality.
- User account settings.

- Power and shutdown options.

The Taskbar provides quick access to frequently used applications.

Users can pin applications to the Taskbar for easier access.

Useful keyboard shortcuts:

Shortcut	Function
Win	Open Start Menu
Win + E	Open File Explorer
Win + R	Open Run dialog
Win + I	Open Settings
Win + D	Show desktop
Win + L	Lock computer
Alt + Tab	Switch applications
Ctrl + Shift + Esc	Open Task Manager

These shortcuts improve productivity when working with Windows.

Task 4 - File Explorer

File Explorer is the Windows utility used to manage files and folders.

It allows users to:

- Create directories.
- Copy and move files.
- Rename files.
- Delete files.
- Search for files.
- Access storage devices.
- View file properties.

Opening File Explorer

Press:

Win + E

Important Windows directories

Directory	Purpose
C:\Windows	Windows operating system files
C:\Program Files	Installed applications
C:\Program Files (x86)	Many 32-bit applications
C:\Users	User profiles and personal files
C:\Users\Public	Shared public user files

C:\Temp	Common temporary-file location, if present

Understanding Windows directory structures is useful when investigating suspicious files and performing forensic analysis.

Task 5 - Windows File System

Windows commonly uses the NTFS file system.

NTFS stands for New Technology File System.

It provides several important features:

- File permissions.
- Access control lists (ACLs).
- File compression.
- Disk quotas.
- File encryption support.
- Journaling.
- Large file and volume support.

Windows also supports other file systems, including FAT32 and exFAT.

Important file extensions

Extension	File type
.exe	Executable program
.dll	Dynamic Link Library
.txt	Text file
.docx	Microsoft Word document
.pdf	PDF document
.ps1	PowerShell script
.bat	Batch script
.msi	Windows Installer package
.sys	System driver file
.log	Log file

From a cybersecurity perspective, file extensions help identify potentially suspicious files, although extensions alone do not guarantee file safety.

Task 6 - Windows Settings

Windows Settings provides a centralized interface for configuring the operating system.

It can be accessed using:

Win + I

Important settings categories include:

- System.
- Bluetooth and devices.
- Network and Internet.
- Personalization.
- Apps.
- Accounts.
- Time and language.
- Privacy and security.
- Windows Update.

These settings allow users to configure their devices and manage system behaviour.

Task 7 - Control Panel

Control Panel is a traditional Windows utility used to configure system settings.

It provides access to various administrative and configuration options.

To open Control Panel:

1. Open the Start Menu.
2. Search for Control Panel.
3. Launch the application.

Some important Control Panel sections include:

- System and Security.
- Network and Internet.
- Hardware and Sound.
- Programs.
- User Accounts.
- Appearance and Personalization.
- Clock and Region.

Control Panel remains useful for accessing legacy Windows configuration options.

Task 8 - Task Manager

Task Manager is a built-in Windows utility used to monitor running processes and system resource usage.

To open Task Manager:

Ctrl + Shift + Esc

Important Task Manager sections:

Section	Description
Processes	Displays running applications and processes

Performance	Shows CPU, memory, disk, and network usage
App history	Displays application resource usage history
Startup apps	Manages applications launched during startup
Users	Displays active user sessions
Details	Provides detailed process information
Services	Displays installed and running services

Task Manager is particularly useful during security investigations because it helps identify unusual processes and unexpected resource consumption.

Task 9 - System Configuration

System Configuration, also known as msconfig, is a Windows utility used to troubleshoot startup and system configuration problems.

To open it:

i. Press Win + R.
ii. Type: msconfig
iii. Press Enter.

Important sections include:

- General - Controls startup selection.
- Boot - Configures operating system boot options.
- Services - Displays Windows services and their startup configuration.
- Startup - Provides access to startup application management in modern Windows versions.
- Tools - Provides shortcuts to additional Windows utilities.

System Configuration is useful when troubleshooting Windows startup problems.

Task 10 - User Account Control (UAC)

User Account Control (UAC) is a Windows security feature designed to prevent unauthorized changes to the operating system.

When an application attempts to perform an administrative action, Windows may display a UAC prompt.

UAC helps protect systems by requiring appropriate authorization before certain administrative operations are performed.

Important concepts:

- Standard user accounts have limited permissions.
- Administrator accounts can perform privileged operations.
- UAC prompts users when elevated permissions are required.

- Administrative privileges should be used carefully.

From a cybersecurity perspective, UAC is an important security boundary-related protection mechanism, although it is not a complete defence against malware.

Task 11 - Windows Command Prompt

Command Prompt is a command-line interface that allows users to execute commands and interact with the operating system.

To open Command Prompt:

i. Open the Start Menu.
ii. Search for cmd.
iii. Launch Command Prompt.

Important commands

Command	Purpose
whoami	Displays the current username
hostname	Displays the computer name
ipconfig	Displays network configuration
ipconfig /all	Displays detailed network information
systeminfo	Displays system information
dir	Lists files and directories
cd	Changes the current directory
mkdir	Creates a directory
type	Displays file contents
tasklist	Lists running processes
net user	Manages or displays local user accounts
netstat -ano	Displays network connections and listening ports

These commands are useful for basic system administration and initial security reconnaissance.

Task 12 - Windows PowerShell

PowerShell is a command-line shell and scripting language developed by Microsoft.
It provides more advanced administration and automation capabilities than traditional Command Prompt.
To open PowerShell:

i. Open the Start Menu.
ii. Search for PowerShell.
iii. Launch Windows PowerShell or PowerShell.

Useful PowerShell commands

Display the current username:

whoami

Display computer information:

Get-ComputerInfo

List running processes:

Get-Process

List services:

Get-Service

Display network configuration:

Get-NetIPConfiguration

Display network connections:

Get-NetTCPConnection

List files and folders:

Get-ChildItem

PowerShell is widely used for Windows administration, automation, and security analysis.

Task 13 - Windows Security

Windows includes several built-in security features designed to protect the operating system and its users.

Important security components include:

a. **Microsoft Defender Antivirus**: Provides protection against malware, viruses, and other malicious software.
b. **Windows Firewall**: Monitors and filters incoming and outgoing network traffic according to configured rules.
c. **Windows Update**: Provides security patches, bug fixes, and operating system updates.
d. **BitLocker**: Provides drive encryption to protect stored data.
e. **User Account Control**: Helps prevent unauthorized administrative changes.

Understanding these features is important when learning Windows security and system hardening.

5. Practical Exercises

Complete the following exercises using the Windows environment provided by TryHackMe.

Exercise 1 - Explore the Desktop

- Examine the Windows desktop.

- Explore the Start Menu.
- Identify the Taskbar and Notification Area.
- Practise using common Windows keyboard shortcuts.

Exercise 2 - Explore File Explorer

- Open File Explorer.
- Navigate through the Windows directory structure.
- Explore the Users directory.
- Examine file properties.
- Identify different file extensions.

Exercise 3 - Explore Task Manager

- Open Task Manager.
- Examine running processes.
- Explore CPU and memory usage.
- Review startup applications.
- Identify running background services.

Exercise 4 - Explore System Configuration

- Launch msconfig.
- Explore the available tabs.
- Examine startup and service configuration options.

Exercise 5 - Explore Command Prompt

Execute the following commands:

```
whoami
hostname
ipconfig
systeminfo
tasklist
netstat -ano
```

Observe the information returned by each command.

Exercise 6 - Explore PowerShell

Execute:

```
Get-Process
Get-Service
Get-ComputerInfo
Get-NetIPConfiguration
Get-ChildItem
```

Understand how PowerShell provides system information and administrative functionality.

Exercise 7 - Explore Windows Security

- Open Windows Security.
- Examine Virus & Threat Protection.
- Explore Firewall & Network Protection.
- Review Device Security.
- Understand the available security settings.

6. Important Windows Shortcuts

Shortcut	Function
Win + E	Open File Explorer
Win + R	Open Run
Win + I	Open Settings
Win + D	Show Desktop
Win + L	Lock Windows
Win + X	Open Quick Link Menu
Alt + Tab	Switch applications
Ctrl + Shift + Esc	Open Task Manager
Win + Shift + S	Take a screenshot
Win + A	Open Quick Settings

7. Key Takeaways

After completing this room, you should understand:

- The basic structure of the Windows operating system.
- How to navigate the Windows desktop.
- How to manage files and folders.
- How Windows uses the NTFS file system.
- How to configure Windows using Settings and Control Panel.
- How to monitor processes using Task Manager.
- How System Configuration helps troubleshoot startup issues.
- How UAC protects administrative operations.
- How to use Command Prompt and PowerShell.
- How built-in Windows security features protect the system.

8. Conclusion

The Windows Basics room introduces the fundamental components of the Windows operating system.

It provides practical knowledge of desktop navigation, file management, system configuration, user permissions, command-line tools, and built-in security features.

These concepts are important for anyone interested in cybersecurity because Windows systems are commonly found in personal computers and enterprise networks.

Understanding the Windows environment provides a strong foundation for learning system administration, Windows security, digital forensics, and penetration testing.